GEBZE TECHNICAL UNIVERSITY

Computer Engineering Department

CSE 396 Module Documentation

Çılgın Kurye

Module M5: Admin Panel and Monitoring

Company: Board&Byte **Group:** Group-8 **Date:** 7 November 2026

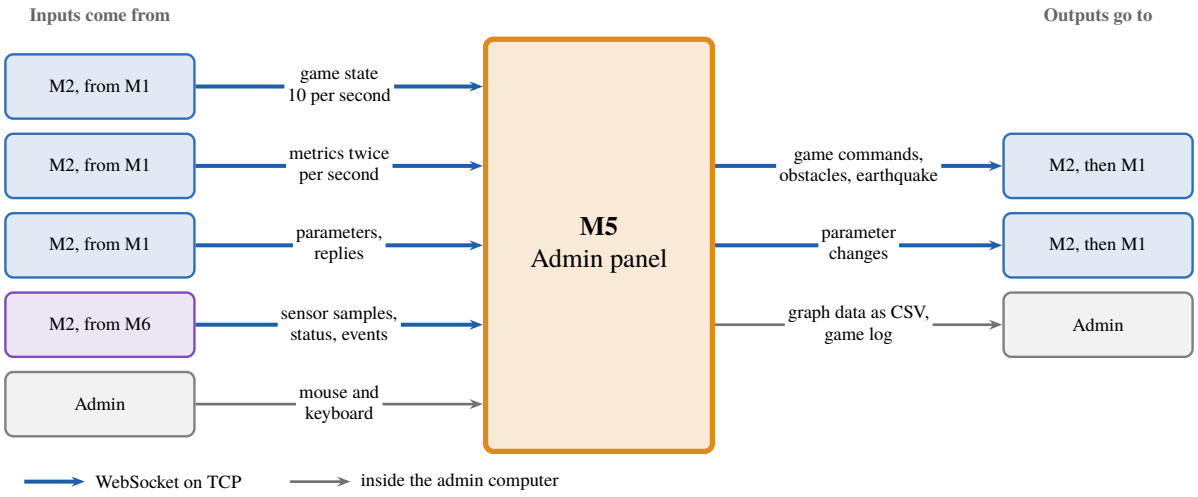
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1. Purpose

M5 is the admin screen. On one page the admin starts and controls the game, changes all parameters during play, puts obstacles on a live map, starts earthquakes and watches the results as live graphs. Each change can be seen in three places: in the 3D city on the phones, on the LED board, and in the graphs on this page.

Runs on	Technology	Requirements
Browser on the main computer, or a laptop on the game Wi-Fi. The page comes from M2	TypeScript web page, a light chart library for the live graphs, a 2D drawing of the city for the live map, MessagePack messages over one WebSocket	R6 R7 R9 R10 R11

The admin page talks only to M2, over one WebSocket. The figure shows what the page gets and what it sends.



2. Requirements covered

M5 takes part in five requirements. The last column says what will work when M5 is finished.

ID	Requirement	What M5 does for it	When M5 is done
R6	Obstacles	Map tools: a brush for each obstacle type, an eraser, a tool to close a road, an obstacle in front of a player, and the traffic density. Parameters for how often each obstacle type appears and how long it stays	The admin can put any obstacle on any road during the game and can change the rate and the duration of each type
R7	Earthquake	An earthquake panel with a strength control and a live seismic graph of the board sensor	The admin can start an earthquake and compare it with a hand hit
R9	Live admin panel	A form for every parameter group, and graphs with a mark for every change	Every change and its effect are seen on one page
R10	Network and measurement	Ping, jitter, loss and frame rate graphs, and the network settings for each player and the board	The admin can make the network worse and measure the result
R11	Engineering requirements	Works against a fake server with recorded data	M5 can be built before the server is ready

3. Inputs and outputs

All data comes through M2. The game state for the map comes less often than to the phones, because the map needs fewer updates.

Direction	Data	From / to	Content
In	Game state	M1, through M2	10 times per second, for the live map
In	Metrics	M1, through M2	Twice per second: scores, delivery times and distances, speeds, falls, traffic, ping, jitter, loss, frame rate, board delay. After a reconnect also the recent history
In	Parameters, replies	M1, through M2	Every parameter with its value and limits, ok or error for each command
In	Board data	M6, through M2	Sensor data, board status, earthquake events
Out	Commands	M2, then M1	Parameter changes, start, pause, resume, end, reset, obstacles, earthquake, new game code
Out	Files	Admin	Graph data as CSV, game log

4. Main components

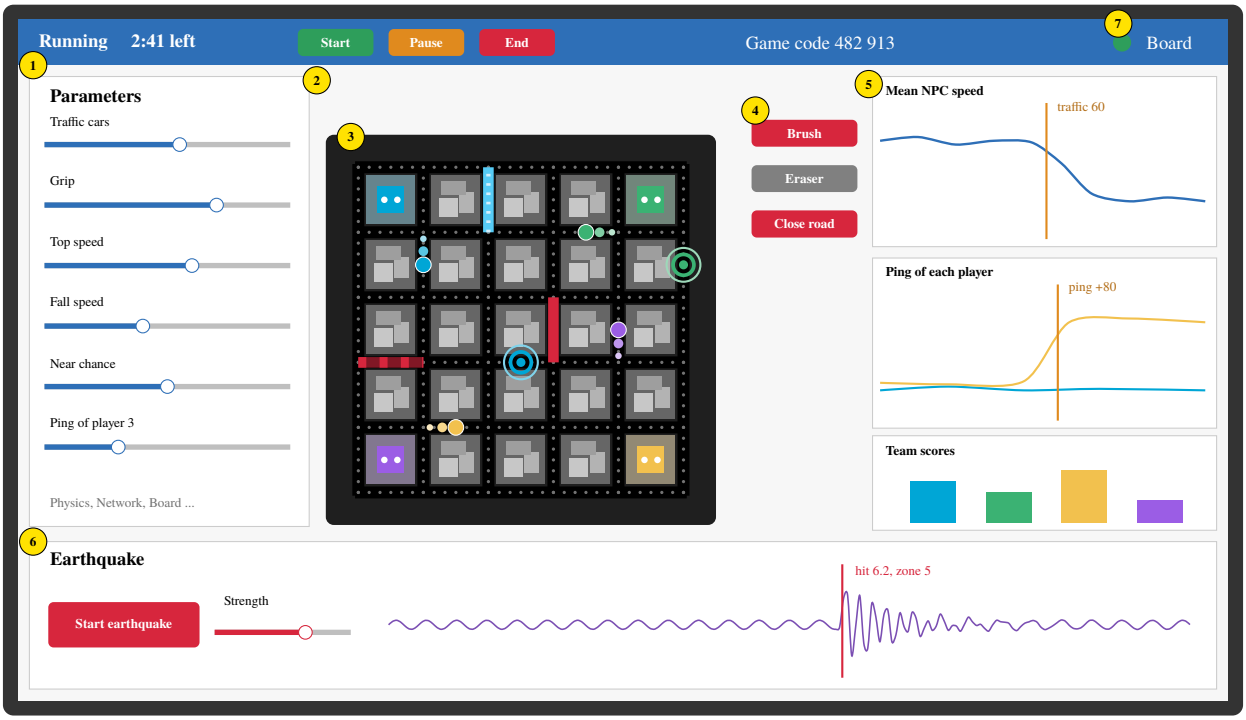
The page is made of five parts. The communication manager is the only part that talks to M2.

Component	Takes	Gives
Communication manager	Messages from M2, commands from the other parts	Decoded data for each part, reconnect and full update after a lost link
Parameter manager	Parameter list with limits, admin input	Forms built from the list, changes sent to M1
Map manager	Game state, clicks on the map	The live map, and obstacle commands with road, lane and position
Graph engine	Metrics, board samples, change marks	Live graphs, the seismic graph, CSV files
Top bar and earthquake panel	Phase, time, board status, button presses	Game commands, earthquake command, the event log

5. How it works

The page:

The figure below shows the planned layout. The top bar shows the game phase, the time left, the buttons to start, pause and end the game, the game code and a small light for the board link (green when the board is connected). On the left are the parameter forms. In the centre is the live map: the city from above, drawn again 10 times per second with all motorcycles, NPC cars, obstacles and delivery points. On the right are the live graphs. At the bottom are the earthquake panel and the seismic graph.



1 Game phase, time and buttons
2 Parameter forms
3 Live map of the city
4 Map tools
5 Live graphs with change marks
6 Earthquake panel and seismic graph
7 Board link light

Parameters:

The parameters are in groups: game, obstacles, players, physics, earthquake, network (one set for each player) and board link. When the page opens, it asks M1 for all parameters with their limits and builds the forms from this list, so a new parameter needs no new page code. The obstacles group has two values for each of the four types: how often it appears by itself and how long it stays.

Network settings:

The network group has one row for each player and one for the board. In each row the admin sets an added delay, a jitter and a loss chance, and for a phone also a frame rate limit. M2 applies these values at once. The ping, jitter and loss graphs show the result next to the change mark.

Cause and effect:

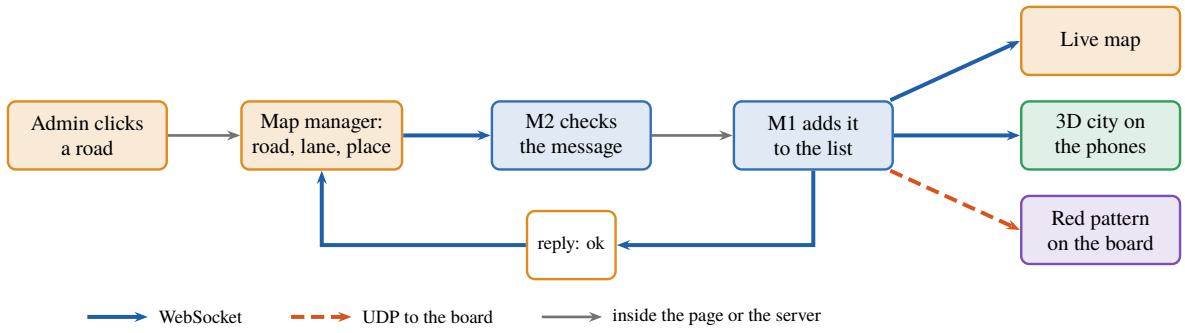
The main goal of the panel is to show the effect of a change. So every change puts a thin vertical line on all graphs, with the name of the parameter. For example, when the admin sets more traffic, the line appears and the graph of the mean NPC speed goes down right after it. At the same time the phones show more cars and the board shows more red. The graph data can be saved as a CSV file for the report.

Adding an obstacle:

The admin first chooses a tool, for example the road work brush. Then the admin clicks on a road on the live map. The map manager finds the nearest road segment, the lane and the position on it with the numbering of the city plan. The communication manager sends an “add obstacle” command with the type, road, lane and position. M1 checks that the road exists. Then it adds the obstacle to the list. In the next tick the obstacle is in the game state, so the map, the phones and the board show it at the same time. It stays for the duration of its type, and then M1 removes it. The eraser removes it earlier with a “remove obstacle” command. “Close road” puts road work on a whole segment, and “in front of a player” puts the chosen obstacle a little ahead of that player’s motorcycle.

Earthquake:

The start button sends a command to M1, and M1 asks the board to run its motors. The seismic graph shows the sensor signal of the board live. Each earthquake event puts a mark on the graph with its strength and zone. A hand hit on the board shows on the same graph, so the admin can compare the two.



After a lost link:

If the WebSocket breaks, the page shows a grey bar and tries to connect again. After the new connection, M2 sends the full parameter list, the current game state and the recent history of all graphs that M1 keeps. So the page fills the forms, the map and the graphs again and the admin can go on at once.

6. Testing

M5 is tested alone against a fake server that plays back a recorded game with its metrics, parameter list and board data. We check every parameter form with good and wrong values, every map tool, all graphs and the CSV export, a second admin login, and that after a lost link the forms, the map and the graphs are complete again.